Saju Khakurel

Embedded Systems Engineer | Sensors, Embedded AI & Industrial IoT

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Profile

Embedded Systems Engineer and Erasmus Mundus master's graduate with nearly three years of professional experience developing embedded C/C++ software for electric vehicles and charging systems. Experienced with STM32, ESP32, sensor integration, firmware development, and hardware testing, with project experience in embedded AI, robotics, and industrial IoT. Currently undertaking a digitalization internship at EJOT.

Professional Experience

Digitalization Intern
EJOT SE & Co. KG
2026/10 – Present | Siegen, Germany
- Supporting digitalization projects within the Digital Competence Center.

Embedded Systems Engineer — Vehicle Software
Yatri Design Studio
2021/12 – 2024/08 | Kathmandu, Nepal
- Developed and tested embedded C/C++ firmware for STM32- and ESP32-based electric vehicle systems, including communication interfaces, diagnostics, telemetry, and device startup.
- Contributed to bootloader and OTA update functions for motorcycles and charging stations.
- Tested firmware on physical hardware using logs, test benches, and debugging tools.
- Developed internal server and OTA functions for 60 kW charging stations and supported testing before deployment.
- Worked with firmware, hardware, backend, and testing teams to find and solve system problems.

Robotics Club Member – ABU Robocon Team
Robotics Club, Pulchowk Campus
2018 – 2020 | Lalitpur, Nepal
- Designed and tested PCBs and embedded firmware for mobile robots.
- Integrated sensors, motors, and encoders and implemented encoder-based motion correction.

Master's Thesis

Comparative Analysis of IIoT Platforms and Implementation of an Open Source Stack-Prototype
University of Siegen, Thesis grade: 1.0
2026/03 – 2026/08 | Siegen, Germany
- Evaluated eight industrial IoT platforms against manufacturing requirements.
- Built and tested four data integration paths using OPC UA, MQTT, ThingsBoard CE, and EdgeX Foundry.

Projects

TinyML Activity Recognition on a Memory-Constrained Smartwatch
- Built a 19-class activity-recognition system using accelerometer data.
- Compared CNN, LSTM, CNN-LSTM, and DeepConvLSTM models and achieved 79.3% Keras test accuracy.
- Applied int8 TensorFlow Lite quantization and reduced the smallest CNN model to approximately 18 KB.

Connected Greenhouse IoT
- Contributed to the integration and testing of an ESP32-S3 and FreeRTOS system with I2C sensors, local display and controls, and an MQTT, Node-RED, InfluxDB, and Grafana monitoring system.

Automatic Modulation Classifier using Deep Learning
- Developed LSTM, Bi-LSTM, and attention-based models for the classification of noisy communication signals.

NAST-Funded Disinfection Robot
Prototype for quarantine areas during COVID-19
- Designed and integrated hardware and embedded software for a mobile robot with sensors, motors, stepper and servo control, pump control, Bluetooth, and Raspberry Pi video.

SLAM and Sensor-Based Perception under ROS
- Contributed to a ROS and TurtleBot3 project using LiDAR data, particle-filter localisation, occupancy-grid mapping, Gazebo, and RViz.

UR5 Cartesian Trajectory Planning & Inverse Kinematics
- Modelled a UR5 manipulator using DH parameters and implemented Cartesian trajectory generation, numerical inverse kinematics, forward-kinematics verification, and 3D motion visualization.
- Compared multiple inverse-kinematics configurations using total joint-motion cost and evaluated trajectory-position accuracy.

Education

Erasmus Mundus Joint Master of Embedded Intelligent Nanosystems Engineering 2024/10 – 2026/09
University of Orléans (France) • University of Siegen (Germany) • Hellenic Mediterranean University (Greece)
- Completed 120 ECTS; final overall grade: 1.6 (German grading scale).
- Specialization in Automation and Robotics at Université d'Orléans.

Bachelor's Degree in Electronics & Communication Engineering 2017 – 2022
Tribhuvan University, Pulchowk Campus Nepal

Skills

Embedded Systems and Hardware (C, C++, STM32, ESP32, Arduino, Raspberry Pi, FreeRTOS, firmware integration, PCB design, test benches, and hardware debugging)

Sensors and Signal Processing (Sensor integration, accelerometer data, I2C sensors, Kalman filtering, DSP, MATLAB, Simulink, and signal preprocessing)

AI and Data (Python, TensorFlow/Keras, TensorFlow Lite, CNN, LSTM, DeepConvLSTM, time-series classification, SQL)

Industrial Communication and IoT (OPC UA, MQTT, REST APIs, CAN, UART, SPI, I2C, BLE, ThingsBoard CE, EdgeX Foundry)

Tools (Git, Linux, Docker, Node-RED, Grafana, InfluxDB, PostgreSQL, KiCad, LTspice, ROS, Gazebo, RViz)

Publications

Book Contribution
- Maharjan, Asim, and Saju Khakurel. "Introduction to IoT." IoT, Machine Learning and Blockchain Technologies for Renewable Energy and Modern Hybrid Power Systems, edited by C. Sharmeela et al., River Publishers, 2022, pp. 1–23

Languages

English C1 | Nepali | Hindi | French A1 | German A1 - basic, currently improving

Volunteering & Leadership

Instructor & Technical Mentor 2019 – 2024
Robotics Club Pulchowk Campus • Locus • Godawari Residential School • Skill Tour Nepal
- Taught Arduino, sensors, motors, PCB design, and embedded programming through practical workshops.
- Mentored student teams working on robotics competitions and science projects.

EV Awareness & Outreach 2024/04 – 2024/05
Yatri Energy / USAID-supported Urja Nepal Project Nepal
- Prepared and delivered technical sessions about electric mobility and charging infrastructure.
- Explained technical topics to students, organisations, and members of the public.